An OCR engine for a historical military archive: two GPUs, five readers and not a single invented word

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Abstract. This report explains how I built the OCR engine that will read the uniform archive of the Museo de Historia Militar: over 300,000 pages of uniform regulations, booklets, official gazettes, army lists, plates and handwritten documents from the 18th to the 20th century. One requirement drives the whole design: the system must not invent anything. Whatever is not read with certainty is returned as `[ilegible]` (Spanish for “illegible”), together with the candidate readings, and it stays that way until two independent readers agree. The engine runs on my own hardware: a server with an RTX 5090 does the reading (PaddleOCR, glm-ocr and qwen3.8), and a second machine with an RTX 5070 Ti cross-reviews with a model from a different family (gemma4). Gemini 3.8 Flash is used only as a last resort, for gaps no one else could resolve and for handwritten documents. During one production night it processed 4,093 pages at about 6.8 s per page. In a test with eight documents submitted together, chaining the two stages cut the time from 451 to 318 s, and Gemini raised the share of resolved `[ilegible]` marks from 46-49% to 87-89%. I also describe what did not work and why each decision changed. **This is a preliminary version:** the system is built and has been tested on about 4,000 real pages, but the main work is still ahead — processing the archive's 300,000+ pages and measuring, at that scale, real accuracy, speed, Gemini cost and human-review volume.

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1. The challenge and its constraints

The Museo de Historia Militar in Málaga is a private collection devoted to Spanish military uniforms. It is currently closed to the public while I look for new premises. This digitisation project makes it possible, in the meantime, to put the archive and the catalogue online, so the documentation can be consulted and searched even while the galleries are closed.

The uniform archive holds more than 300,000 digitised pages from very different sources. There are printed regulations (royal orders, circulars, regulations and uniform booklets), serial official publications (Diario Oficial, Gaceta, Colección Legislativa, army lists), books and magazines, plates and photographs, and handwritten archival documents. The typography spans the 18th to the 20th century, the paper is often yellowed or the ink faded, and tables of amounts, troop strengths or garments are everywhere.

From each document the uniform catalogue needs four things: the text with the coordinates of every line, the page regions (text, titles, tables, figures, stamps), the tables in a usable structure, and the document type. Five constraints shaped every decision.

- **Do not invent.** Plausible but false text is worse than a gap: it enters the catalogue and nobody looks at it again. A vision model that “fills in” a word it cannot see, or a year that is not there, contaminates the catalogue without a trace. So everything that is not read with certainty is returned marked as `[ilegible]`.
- **Own hardware.** Reading is done locally, on a server and a workstation I already had. The server runs no service that connects outward; the only exception, added later and deliberately, is Gemini as a last resort (section 3.6).
- **A single entry point.** The catalogue talks only to one protected HTTPS entry point. How the work is split between machines is invisible to it.
- **Volume.** At the archive's scale, one second per page is 3.5 days of continuous processing. Speed matters as much as quality.
- **Faithfulness to the original.** Spelling is not modernised and abbreviations are not expanded: «Reales ordenes», «paño», «dho.», «112'50» and «Idem» are transcribed as they are.

2. System overview

The system is an orchestrator (FastAPI) that hands each document to five readers with different roles, in two chained stages (Figure 1). The first stage runs entirely on the server: page analysis with PaddleOCR, classification with qwen3.8, reading of tables and figures, cross-checking doubtful lines with glm-ocr, and a first resolution of `[ilegible]` marks with qwen3.8. The second stage starts as soon as the first ends, while the server is already working on the next document. In it, gemma4, on the second machine, independently reviews what is doubtful; then Gemini resolves what nobody else could, and what remains goes to the dictionary for human review.

The whole system, from input to catalogue

Each document goes through two stages on two different machines; manuscripts take a shortcut.

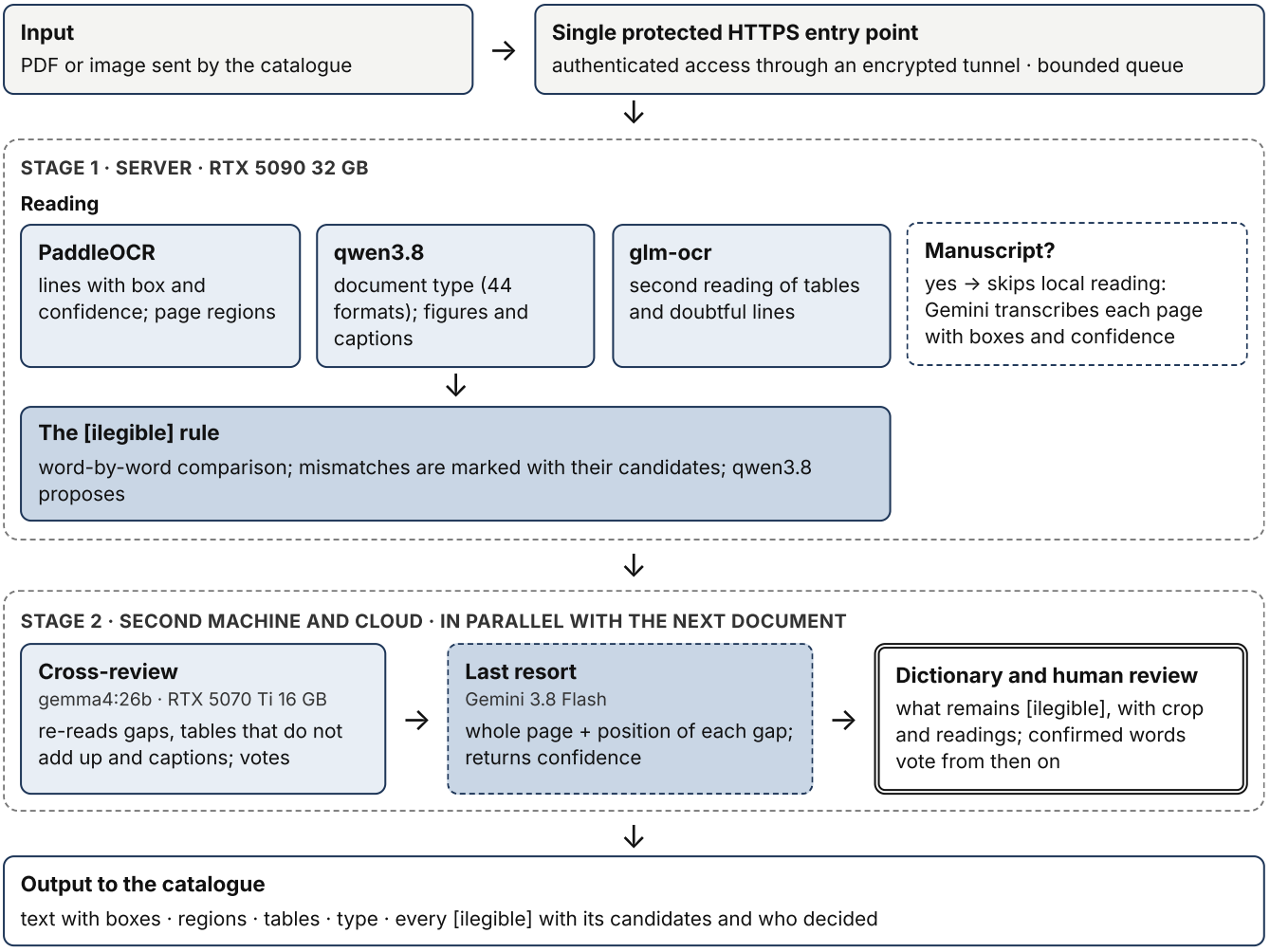


Figure 1. Stage 1 uses only the server's GPU; stage 2 uses the second machine's GPU and the Gemini API, so they can run in parallel without getting in each other's way. Dashed outline: optional or external steps; double outline: human involvement.

Three ideas run through the whole design:

- **Independent readers from different model families.** PaddleOCR (Baidu), glm-ocr (Zhipu), qwen3.8 (Alibaba), gemma4 (Google, local) and Gemini (Google, cloud) make different kinds of mistakes. When two agree, the reading is far more reliable than any one of them alone.
- **Better a gap than an error.** When in doubt, the system marks and explains: every `[ilegible]` carries the readings of all engines and the decision taken.
- **Check against what the document says about itself.** In tables, the “Suma” and “Total” rows make it possible to verify the figures. A correction is only accepted if it makes the sum add up.

2.1 Hardware and software

Table 1. Machines and engines.

Component	Detail	Role
Server	Linux, NVIDIA RTX 5090 (32 GB), Ollama	Stage 1 and orchestrator
Second machine	Windows, NVIDIA RTX 5070 Ti (16 GB), Ollama	Cross-review (stage 2)
PaddleOCR	PP-OCRv6 medium (detection and recognition), PP-DocLayout_plus-L, TableRecognitionPipelineV2	Lines with box and confidence, regions, tables
glm-ocr	0.9 B parameters	Second reading of lines and tables
qwen3.8	27 B parameters, about 96 tokens/s	Classification, figures, first resolution
gemma4:26b	Mixture of experts; about 82 tokens/s once loaded	Independent review
Gemini 3.8 Flash	Google API, low thinking, 6 requests in parallel	Last resort and manuscripts

Before installing anything I cleaned up the server: I removed services that were not needed and cut automatic outbound connections (unattended updates, reporting, telemetry). The orchestrator starts with the models in offline mode. The only route to the Internet is the Gemini API, which is switched on or off in the configuration.

2.2 A single entry point

The catalogue sends each document to a single protected HTTPS entry point, with authenticated access through an encrypted tunnel. The orchestrator exposes no port to the Internet and is the only component that talks to the second machine, over the internal network. So the catalogue does not know, and does not need to know, how many machines are behind it: if I add a third tomorrow, nothing changes for it.

3. The process, step by step

3.1 Reading and page analysis

Each PDF page is rendered at 200 dpi, except large-format pages (plates, plans), which are capped at 5,000 px per side. PaddleOCR returns each line with its text, confidence and box, and PP-DocLayout classifies the page regions. I chose PaddleOCR as the base reading for three reasons: it gives precise line boxes (which generative vision models do not guarantee), it is fast (0.1 to 0.5 s per page), and its per-line confidence is a useful signal for deciding what deserves a second reading.

At first, Paddle's table reader used an older recognition model (PP-OCRv4) that dropped accents, and it took 52 s to load the first time. I fixed this by using the same v6 models in every phase and loading them when the service starts.

3.2 Document classification

qwen3.8 classifies the whole document, not just the cover. It sees three sample pages (the cover, the page with most text and the page with the largest figure) together with the OCR

text and the region counts for the whole document. At first it classified from the cover alone, and a regulation with plates came out as “plate”.

The 44 types describe the format, not the subject, and match those of the catalogue. Besides the main type, the classifier lists the types the document contains: a Diario Oficial that publishes a royal order with a plate is “official gazette” and contains “royal order” and “plate”. Each type comes with a short definition, because some are close to each other (commission and warrant, report and study, internal bulletin and Gaceta). The classification also flags whether the document is mostly handwritten, which is independent of the type: a service record can be printed or written by hand.

3.3 The `[ilegible]` rule

This is the core of the system (Figure 2). A line that PaddleOCR reads with low confidence (below 0.85) is cross-checked with glm-ocr and compared word by word. Accents, spaces, case, punctuation and the form of ordinals (`3.a`, `3.ª`, `3.º`, `2."`) do not count as differences. Any different letter or digit turns that span into `[ilegible]`, and both readings are kept — for example «Reqimiento» vs «Regimiento», or «14» vs «11».

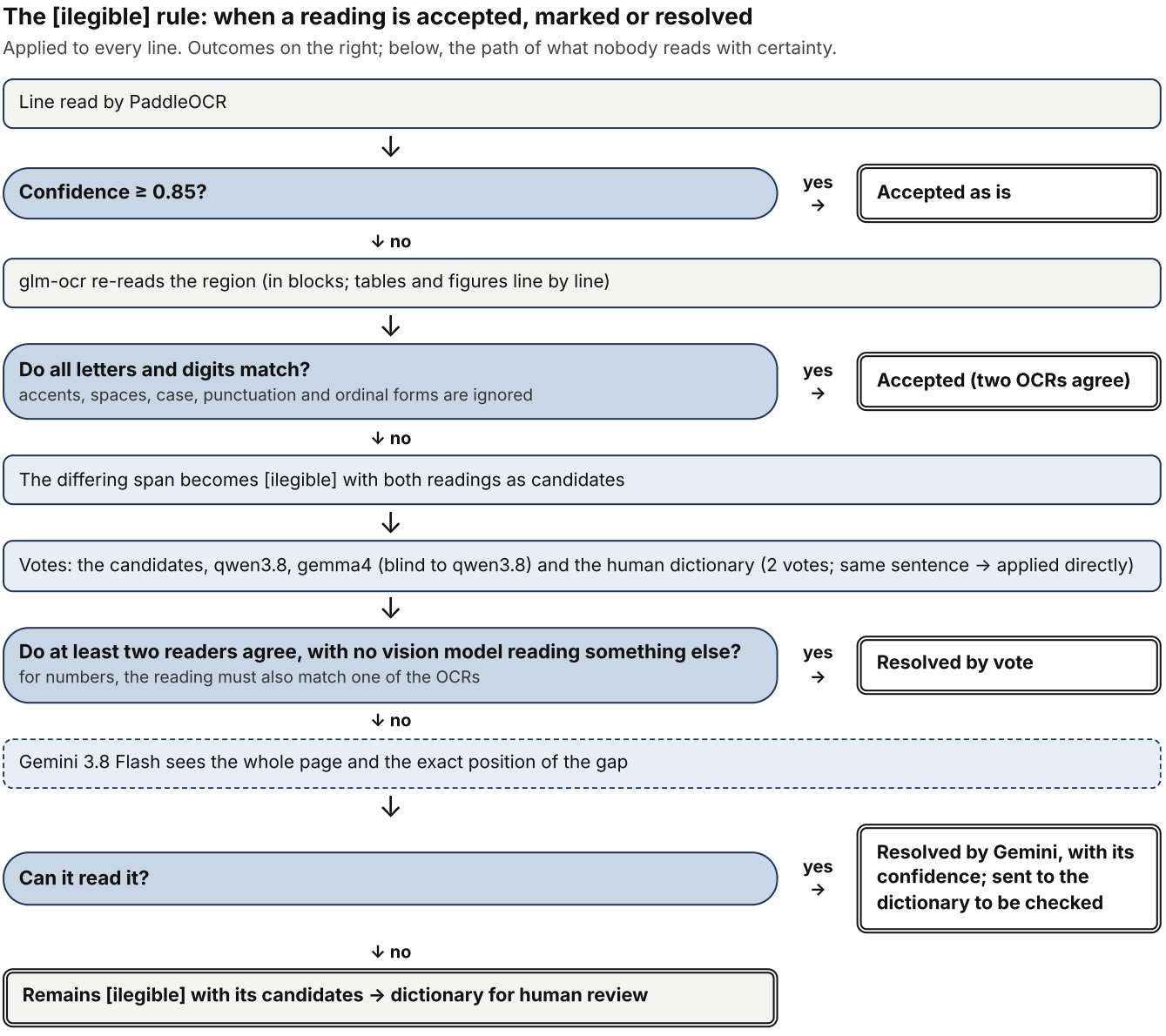


Figure 2. The decision for each line. Outcomes with a double outline are the ones that reach the catalogue; Gemini (dashed) can be switched off, in which case anything unresolved goes straight to the dictionary.

Next, qwen3.8 looks at the line crop, with the candidate readings, and proposes the word only if it reads it with certainty. Two rules limit what it can do:

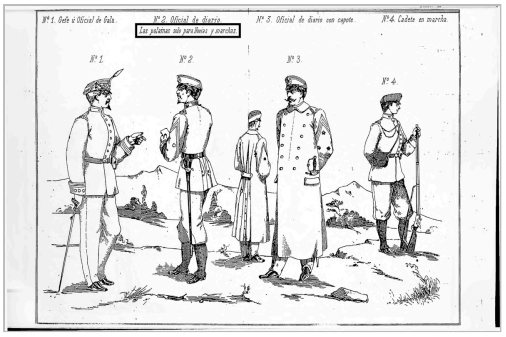
- **It does not touch what was already certain.** Its answer is rejected if it changes the parts of the line on which both OCRs agreed.
- **No third readings for numbers.** A year, a date or an amount must match what one of the OCRs read. This rule came from a real case: on the title page of the 1867 booklet, Paddle read «157.1.» and glm-ocr «1871.», and in an intermediate version qwen3.8 proposed «1571». Today the year stays marked, with its two candidates.

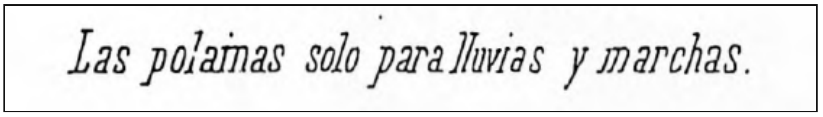
qwen3.8's reading is never accepted on its own. Each gap is decided by a vote among the readers: the two OCRs, qwen3.8, gemma4 (section 3.5) and the human dictionary (section 3.8). The reading backed by at least two readers is accepted, provided no vision model gives a different one. Every `[ilegible]` in the result carries its candidates, each model's reading, Gemini's confidence if it was involved, and the decision taken, so the catalogue can explain every word: who read it and why it was accepted.

Figure 3 follows a real gap from start to finish. In version 1.4, glm-ocr and qwen3.8 made the same mistake («paralluvas»), and gemma4 stopped it: the gap stayed marked with its candidates instead of entering the catalogue as a word that does not exist.

A real gap, before and after

Infantry uniform regulations booklet (cartilla, 1867-1874), plate on p. 67: «Las polainas solo para lluvias y marchas.» (“Gaiters only for rain and marches.”)





Reader	Reading
PaddleOCR (confidence 0.83)	`Las polanas solo parauvias ymarchas.`
glm-ocr	`Las polainas solo paralluvas y marchas.`
Comparison	`Las [ilegible]¹ solo [ilegible]²` ¹ polanas / polainas ² parauvias ymarchas. / paralluvas y marchas.
Gap ¹	qwen3.8 and gemma4 read «polainas» → resolved
Gap ² · version 1.4	qwen3.8: «paralluvas y marchas.» · gemma4: «para lluvias y marchas.» → they disagree → stays [ilegible] with its candidates
Gap ² · version 1.7	qwen3.8 and gemma4 read «para lluvias y marchas.» → resolved

Result with version 1.4

`Las polainas solo [ilegible]`

candidates: parauvias ymarchas. · paralluvas y marchas.

Result with version 1.7

`Las polainas solo para lluvias y marchas.`

> glm-ocr and qwen3.8 made the same mistake in 1.4; without a reader from another model family, «paralluvas» would have been accepted. In 1.7, qwen3.8 gets it right, gemma4 agrees and Gemini is not needed.

Figure 3. Actual system output during the production night (version 1.4) and when the page was reprocessed with the current version (1.7). An 1867 print with no personal data.

Doubtful lines are cross-checked in blocks: glm-ocr reads the whole region at once (up to 30 lines) and its reading is aligned word by word with Paddle's lines. Lines in tables, figures and stamps are checked one at a time, because glm-ocr, reading a table of figures as running text, went into loops (section 7).

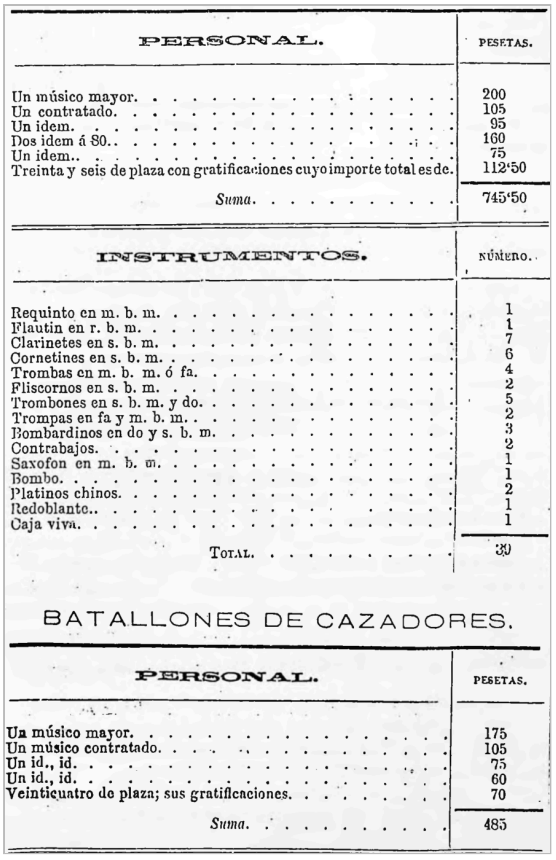
3.4 Tables: double reading and sums

Each table is read with glm-ocr, which reconstructs structure best, and converted to HTML. PaddleOCR re-reads it only if glm-ocr fails, if its text drifts from the OCR, or if its sums do not add up. Sometimes glm-ocr misses the bottom of a region (two tables in a row, for instance): the OCR lines that do not appear in any cell are then detected, the missing part is cropped and read again.

Then the “Suma”, “Total” and similar rows are checked against the sum of their section. The parser understands `112'50`, `745,50`, `1.250,75` and the old thousands sign (`2U500`). A cell with «ídem» or ditto marks counts as the value in the row above. If a sum does not add up, the cells in that column are re-read in stage 2, first by gemma4 and, if it still does not add up, by Gemini. A correction is only accepted if it makes the sum add up (Figure 4).

Checking sums: fix misreadings, do not “fix” the original

Infantry uniform regulations booklet (1867-1874), p. 55. Three sum checks on the same page.



PERSONAL.	PESETAS.
Un músico mayor.	200
Un contratado.	105
Un idem.	95
Dos idem á 80.	160
Un idem.	75
Treinta y seis de plaza con gratificaciones cuyo importe total es de.	112'50
Suma.	745'50

INSTRUMENTOS.	NÚMERO.
Requinto en m. b. m.	1
Flautin en r. b. m.	1
Clarinetes en s. b. m.	7
Cornetines en s. b. m.	6
Trombas en m. b. m. ó fa.	4
Fliscornos en s. b. m.	2
Trombones en s. b. m. y do.	5
Trompas en fa y m. b. m.	2
Bombardinos en do y s. b. m.	3
Contrabajos.	2
Saxofon en m. b. m.	1
Bombo.	1
Platinos chinos.	2
Redoblante.	1
Caja viva.	1
TOTAL.	39

BATALLONES DE CAZADORES.

PERSONAL.	PESETAS.
Un músico mayor.	175
Un músico contratado.	105
Un id., id.	75
Un id., id.	60
Veinticuatro de plaza; sus gratificaciones.	70
Suma.	485

Row	Read	Check	Outcome
Sum (personnel)	745'50	rows add up to 747.50	**error in the printed original**: flagged; gemma4 and Gemini review it and leave it
Total (instruments)	30	rows add up to 39	gemma4 re-reads the faded cell: **39**; now it adds up
Sum (Cazadores)	485	adds up	second table on the page, recovered in full

> A mismatch is not always a reading error: the print says 745'50 where the rows add up to 747.50. The system flags it for human review.

Figure 4. The three sum checks on page 55 and what came out of each.

A table with more than three mismatches is not re-read cell by cell. It is usually a misread structure, or “totals” that are not sums, such as the strength tables of an army list. It is flagged for human review.

3.5 Cross-review on the second machine

The second machine runs gemma4:26b, a model from a different family than qwen3.8. It receives the same crops and the same OCR readings, but *not* qwen3.8's proposal, so that its vote is independent. It reviews three things: `[ilegible]` gaps, tables whose sums do not add up, and figure captions the OCR did not back. The voting rule is the one in section 3.3; if gemma4 and qwen3.8 read different things, the gap stays marked even if one of them matches an OCR.

I chose it because it is the best vision model available in Ollama that runs on a 16 GB GPU at a useful speed: about 0.7 s per reviewed line. During the production night, gemma4 confirmed 1,961 readings, resolved 1,314 on its own and stopped 973 that qwen3.8 had accepted and no other reader backed. If the second machine is off, the document is still processed, without this review, and the result says so.

3.6 Gemini as last resort

Whatever is still unresolved after four readers is sent to Gemini 3.8 Flash, in a single call per page. Each call carries the whole page and, for each gap:

- the box of its line, in Gemini's format: `[ymin, xmin, ymax, xmax]` normalised to 0-1000;

- the approximate position of the word within the line;
- the line with the gap marked and the neighbouring lines, as context;
- what each engine read.

Gemini answers all the gaps on the page at once and states its confidence in each. Since it is reached only when nobody else could resolve the gap, its reading is accepted. The confidence is stored in the result and in the dictionary, so a person can review the lowest-confidence ones first. Pages are sent six at a time. With thinking at the default level, each page took 5 to 90 s; with low thinking, about 5 s, which adds 4 to 8 s per 12-page document.

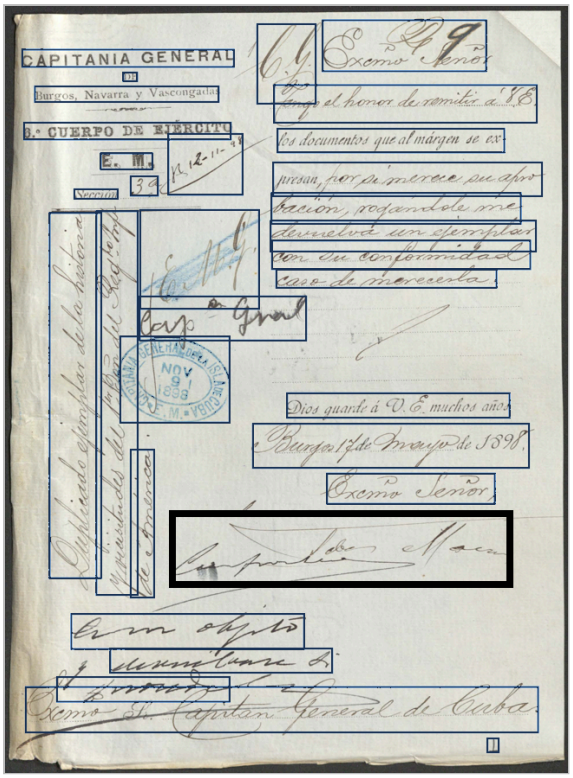
Gemini is the server's only outside connection, and that was a deliberate decision: pages that reach this phase leave for Google. It can be disabled per document or for the whole service.

3.7 Manuscripts

A mostly handwritten document cannot be read well with the local engines. Before separating them, a 23-page file left 2,184 `[ilegible]` marks after going through every reader. Now, when the classifier flags a document as handwritten, the server only does page analysis and classification, and Gemini transcribes each whole page. It returns the lines in reading order, each with its box, the literal text (with `[ilegible]` where it cannot read) and its confidence (Figure 5).

Manuscripts: direct transcription with Gemini

1898 file (Captaincy General of Burgos, Navarre and the Basque Provinces), p. 1. Boxes returned by Gemini: thin line, confidence ≥ 0.8; thick line, < 0.8.



Text	Conf.
CAPITANIA GENERAL / DE / Burgos, Navarra y Vascongadas	0.98-0.99
6.º CUERPO DE EJÉRCITO · E. M. · Sección 3.ª	0.97-0.99
Excmo. Señor	0.96
Tengo el honor de remitir á V.E.	0.98
los documentos que al márgen se ex-	0.98
presan, por si merece su apro-	0.97
Duplicado ejemplar de la historia (margen)	0.95
y vicisitudes del 1.er Bon del Reg.to Inf.ª	0.95
CAPITANIA GENERAL DE LA ISLA DE CUBA / NOV 9 1898 / E. M. (stamp)	0.92
Burgos 17 de Mayo de 1898.	0.98
Castor de Moxin (signature)	**0.75**
Excmo. Sr. Capitan General de Cuba.	0.97

> The only line below 0.8 is the signature. Over 6 pages: 180 lines in 44 s, mean confidence 0.96. Before, a 23-page handwritten file left 2,184 [ilegible] marks.

Figure 5. 1898 file, page 1, with the line boxes returned by Gemini. The people named in the document are from the 19th century.

3.8 Dictionary and human review

Every remaining `[ilegible]` goes into a dictionary for human review. Each entry holds:

- the word (`[ilegible]` until someone reads it);

- the sentence it appears in;
- each engine's reading;
- a crop of the line, and the source document and page.

A review web page shows the pending entries with their crops. It is enough to type the word or click one of the proposed readings. Gemini's resolutions are listed too, from lowest to highest confidence, so that a person can confirm or correct them.

What a person confirms becomes one more reader. If the sentence is the same (a document processed again), the reading is applied directly. If it is the same OCR confusion in another sentence, it counts as two votes, unless a vision model reads something else. If two people validated different words for the same confusion, the dictionary is not used for that case. After the first production night, the dictionary held about 3,000 pending entries.

3.9 Period signs and conventions

These documents use signs that are neither errors nor illegible text and that, until they were explained to the correctors, produced false marks (Table 2). The three correctors (qwen3.8, gemma4 and Gemini) receive the guide with every request. The same guide feeds the comparison between readings and the table arithmetic. On seven pages of the 1867 booklet, the marks dropped from about 15 to 9; the ones left are real errors, such as «FigunA» vs «FIGURA».

Table 2. Conventions the system recognises.

What	Forms	Meaning and handling
Repetition	Ídem, ídem, id., íd.; quotes or signs on their own: " ″ » « >> << 〃	“Same as above”. The sign is transcribed; in tables it counts as the value in the row above and is flagged in the result.
Ordinals	1.º, 2.ª, N.º (OCR reads them as 1." or 2.a)	«2."» and «2.º» are the same reading.
Abbreviations	D., D.ª, S.M., V.E., Excmo., R.O., R.O.C., D.O., C.L., dho., qe., Tte., Cap., Rgto., Inf.ª, 7bre… Xbre	Transcribed without expansion.
Currencies and measures	rs. vn., mrs., escs., ptas., cts., @ (arroba), lib., onz., fan., vs., pulg.	Ignored when reading the number.
Numbers	112'50; 2U500	112.50; 2,500 (old thousands sign).
Empty cell	— · .... · …	“Nothing” or “not applicable”; does not break the table section.

4. Operations: queue, stages and logs

The system has to withstand months of continuous load without losing jobs or becoming saturated. These are the operational decisions:

- **Bounded queue.** Up to 200 jobs can wait; beyond that, the catalogue is asked to retry later. If free disk space runs low, no new jobs are accepted. Each accepted job returns its position and an estimated wait based on how long real jobs take.
- **Two stages in parallel.** The server processes one document while stage 2 finishes the previous one. At most one document waits between them, so there are never more than three documents in memory.

- **Restarts without loss.** On start-up, queued jobs, and the one in progress, resume automatically. I have checked this several times with real jobs.
- **GPU memory.** PaddleOCR releases its memory after each page. If it still runs out, the Ollama models are unloaded and the page is retried; if it fails again, it is retried at a reduced size.
- **Logs.** Each job leaves one line with times per phase, pages, type, `[ilegible]` marked and resolved, tables and each reader's results. The text is not stored. Each exception is logged with its trace and a copy of the document that caused it, so it can be reproduced. Every hour, a report summarises the previous hour.
- **Retention.** The uploaded document is deleted once processed, and the result after 4 hours or once downloaded. Copies of documents that caused exceptions are kept 30 days; logs, 90 days. The second machine stores nothing.

5. How I tested it

I do not yet have a set of hand-transcribed pages to measure character-level accuracy. What has been measured so far are internal signals: how many lines both OCRs read the same way, how many gaps are marked and resolved, how many sums add up, and how long each phase takes. On top of that, I reviewed the difficult cases by hand. This is the main limitation of this report (section 8).

Table 3. Test sets.

Test	Content	Purpose
1867 booklet	Infantry uniform booklet, 72 pages; tables on pp. 55-56, plates on pp. 63-67	Fixed reference across versions; every mark reviewed by hand
106 samples	Two pages of each document, 11 catalogue types	Speed and robustness (v1.1)
Batch of 8	Army lists, 1826 Estado Militar, 1929 Colección Legislativa, books, booklets; 12 pages each, submitted together	Comparing versions with a full queue
Production night	149 archive documents, 4,093 pages in about 8 hours	Behaviour under real load
Manuscript	1898 file from the Captaincy General of Burgos	Manuscript route

6. Results

6.1 Version history

The system reached its current form in seven versions. Each one responded to a problem measured in the previous one (Figure 6).

Version history

What changed in each version and what was measured.

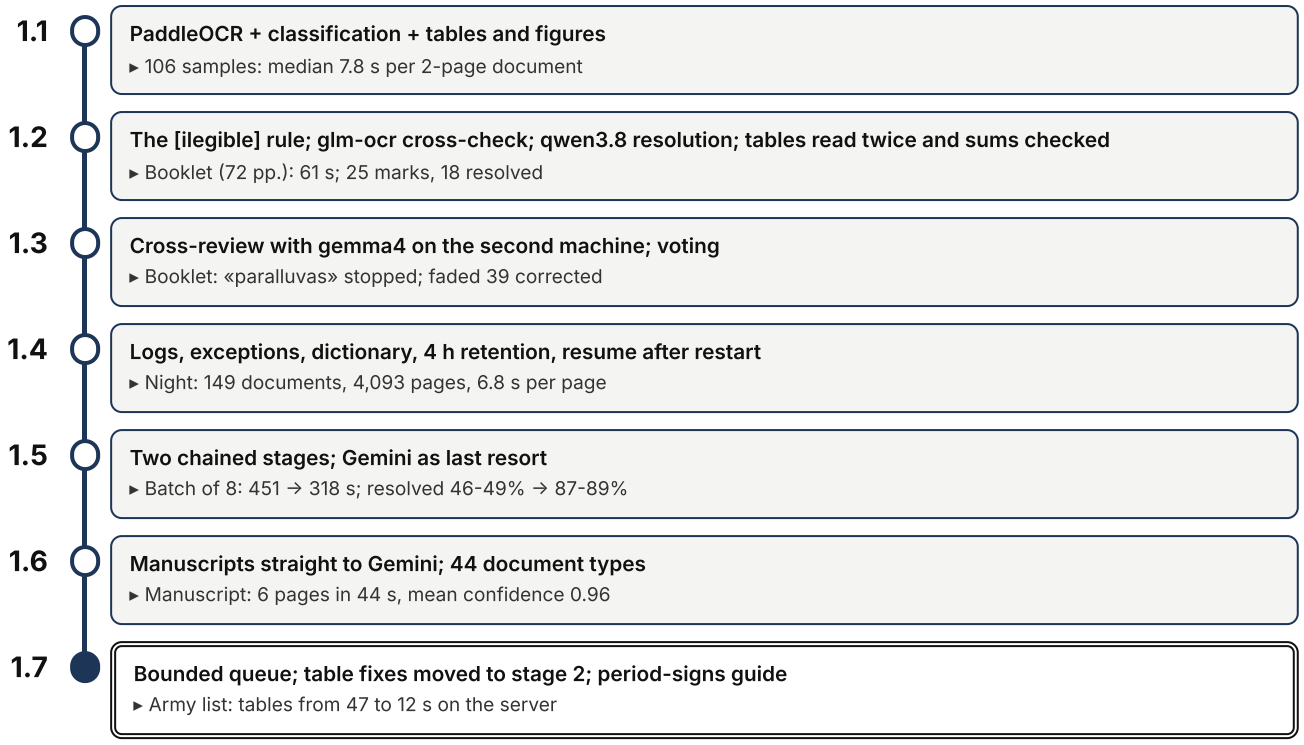


Figure 6. How the system evolved. Version 1.7 is the current one.

6.2 The production night

During a production night of about 8 hours, the catalogue sent 149 documents without a break, with version 1.4 (a single stage, review capped at 80 items per document, no Gemini).

4,093 pages in about 8 hours	**6.8 s** per page (about 500 pages/hour)	**511,017** lines read; 96% with no doubt	**17** failed documents (11%), all from GPU memory

Seconds per page in each phase (production night)

132 completed documents, 4,093 pages, version 1.4. Each value is the phase time divided by the document's pages.

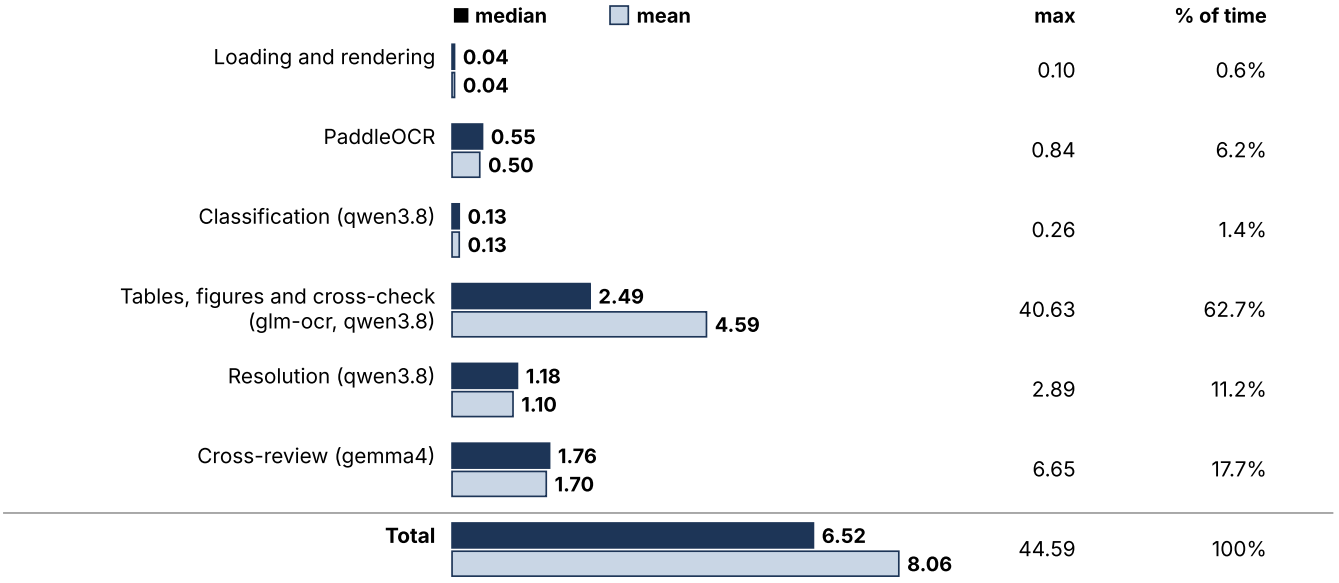
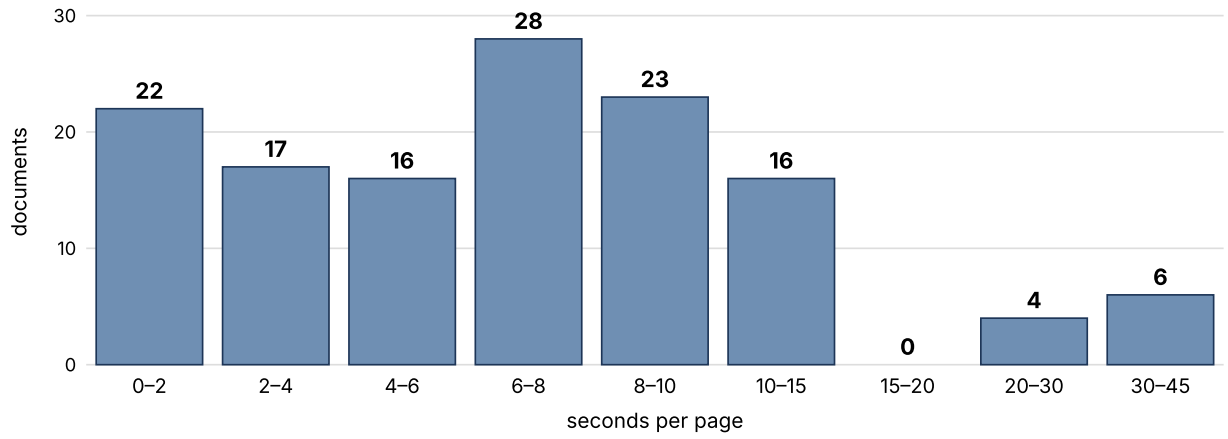


Figure 7. Time per page for each phase. The tables, figures and cross-check block takes 63% of the time and varies most from document to document (up to 40.6 s per page). The finer breakdown in version 1.7 showed that, within that block, the expensive part was not the line cross-check (0.3 to 5 s per 12-page document) but re-reading cells whose sums did not add up, regions taken for handwriting, and figures.

Distribution of seconds per page per document

Production night: 132 completed documents. Each bar counts documents by their mean time per page.



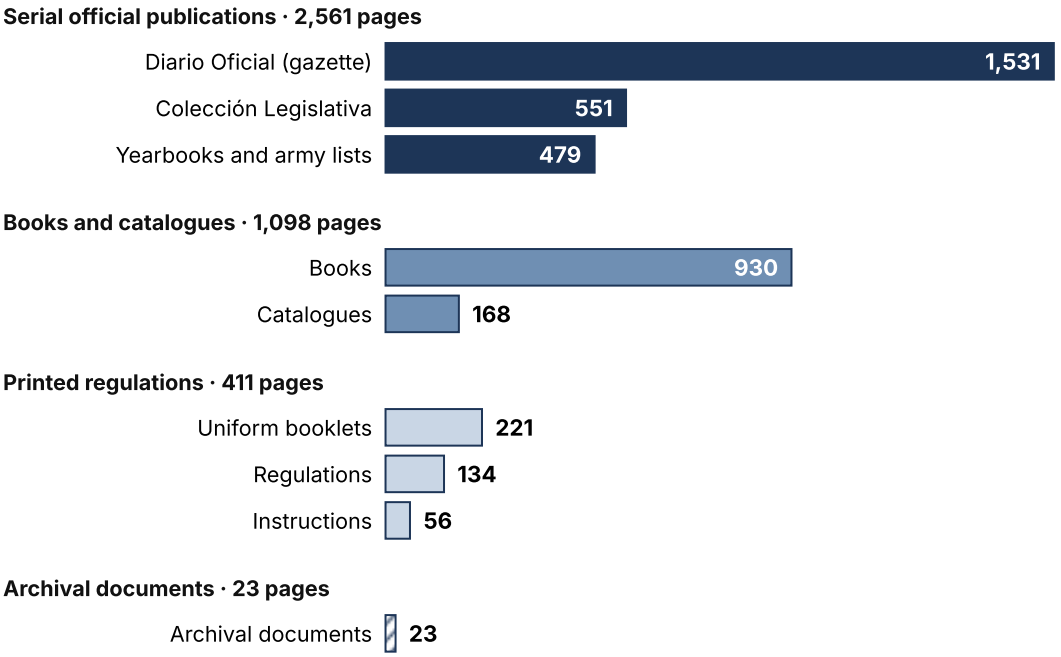
Median 6.5 s; page-weighted mean 6.8 s. The 10 documents above 20 s/page are 4 official gazettes and 6 volumes of the Colección Legislativa: in all of them over 80% of the time went to tables, figures and line cross-checks, with 41 to 457 [ilegible] marks each.

Figure 8. The logs record time per document, not per page, so the histogram uses each document's mean.

Of the 19,757 `[ilegible]` marks, 3,276 were resolved (17%). The figure is low for two reasons. First, they are highly concentrated: ten documents hold 54% of the pending ones, and they are very poor scans or manuscripts, with up to 92 marks per page. Second, the review was capped: in the 80 documents gemma4 could review in full, 42% were resolved, and in the 52 that hit the cap, 11%. Both problems are solved: manuscripts now go to Gemini (section 3.7) and the review has no cap. In tables, the sum checks corrected 505 cells, and 225 of 1,757 tables (13%) were left with a mismatch for review.

Pages by detected document type

The 4,093 pages processed during the production night, about 8 hours (132 documents), as classified. This is that night's sample, not the whole archive.



> 1,757 tables were detected in those pages. The classifier is not validated yet; that version did not flag manuscripts, and there were no loose plates in the batch.

Figure 9. That night's pages by detected type. It is there to help read Figures 7 and 8, not as an inventory of the archive.

6.3 Chaining stages and adding Gemini

With version 1.5, the same batch of eight 12-page documents was processed with chained stages and then with Gemini as last resort (Figures 10 and 11).

[ilegible] resolved: local readers only vs adding Gemini

Batch of 8 documents of 12 pages (army lists, Estado Militar, Colección Legislativa, books, booklets).

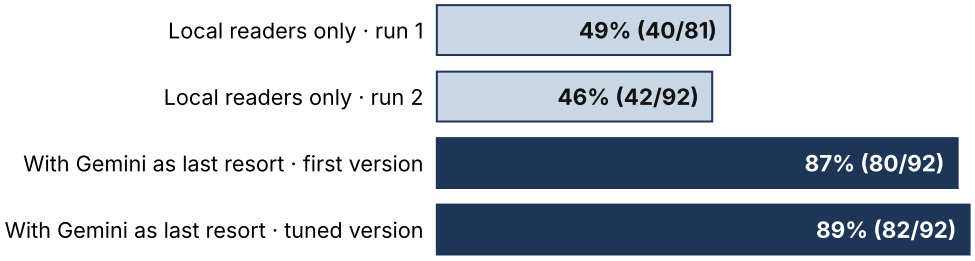


Figure 10. Share of `[ilegible]` marks resolved in the batch. In brackets, resolved over marked in each run.

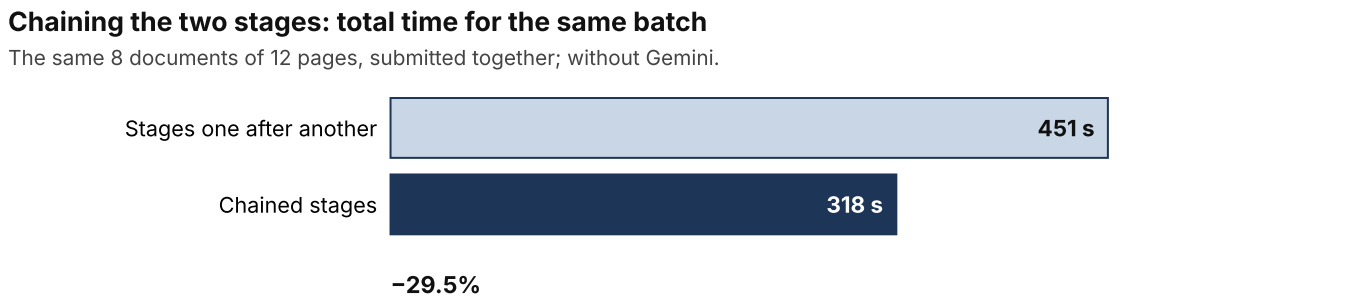


Figure 11. Wall-clock time for the whole batch. In the first version with Gemini, which used default thinking and sent pages one at a time (5 to 90 s each), the batch took 780 s; with low thinking and 6 pages in parallel, Gemini adds 4 to 8 s per document.

In the batch, the server (stage 1) accounts for about 280 s and stage 2 for about 200. Chained, the total approaches the slower stage rather than the sum. Today the bottleneck is the server, and within it, tables.

6.4 Tables and manuscripts

Table 4. Measured changes on a 12-page army list and on the manuscript.

Case	Before	After	What changed
Army list tables, on the server	47 s	12 s	Cell re-reading moved to stage 2. The strength table (32 “mismatches” that are not sums) is flagged for human review without attempting it.
Army list, stage 1	≈ 70 s	≈ 40 s	The above, plus PaddleOCR only when glm-ocr fails (2 s less).
Gemini on the army list tables	> 10 min	4 s	Before, one call for every sum that did not add up. Now at most 3 per table, in parallel.
Handwritten file	2,184 marks in 23 pp.	6 pp. in 44 s	Straight to Gemini; mean confidence 0.96.
“Handwritten” areas in prints	30-52 s	0 s	No longer re-read (a signature was enough to re-read the whole text).

7. What worked and what did not

Figure 12 summarises the decisions with the measurement behind each one. The three main lessons: marking instead of guessing prevents errors nobody would look at again; a reader from another model family catches errors two engines share; and table sums are the only objective evidence that does not depend on any model. Almost everything that failed had the same root: asking a model to read with no reference to check it against, or paying a fixed cost (memory, re-reading, thinking) on documents that did not need it.

What worked and what did not, with the measurement that decided

Every decision was taken from a measurement on real documents.

Worked

	measurement
✓ **Marking instead of guessing**	avoided the invented year «1571» on a title page
✓ **Readers from different model families**	gemma4 stopped 973 readings backed only by qwen3.8
✓ **Sums as objective evidence**	505 cells corrected in one night; 745'50 kept as an error in the print
✓ **Two stages on two GPUs**	451 → 318 s on the same batch
✓ **Gemini only at the end**	46-49% → 87-89% resolved; 4-8 s per document
✓ **Manuscripts straight to Gemini**	2,184 marks in 23 pp. → 6 pp. in 44 s
✓ **Logs with a copy of the file**	cause of 17 failures found in minutes

Did not work → what we did

	measurement	decision
× **Vision model as a free second opinion**	loops and plausible but false readings	→ glm-ocr + word-by-word comparison
× **Describing every figure**	objects invented at scan edges	→ edge filter; “text” or “empty” answer allowed
× **Classifying from the cover**	regulations with a cover plate taken as plates	→ 3 sample pages + regions of the whole document
× **Loose comparison**	«Reqimiento» passed as correct	→ skeleton comparison
× **qwen3.8 resolving alone**	invented «1571»	→ digit rule + voting
× **Cap of 80 review items**	11% resolved in documents that hit it	→ no cap
× **GPU memory held by PaddleOCR**	17 of 149 documents failed (11%)	→ release after each page; retries
× **glm-ocr on the second machine**	does not fit next to gemma4 in 16 GB	→ stays on the server
× **glm-ocr in blocks to gain speed**	no measurable gain (0.3-5 s per document)	→ kept: it now checks every doubtful line
× **Blocks including table lines**	glm-ocr looping (14 cases)	→ tables and figures line by line
× **Re-reading “handwritten” areas of prints**	30-52 s per document	→ removed; full manuscripts go to Gemini
× **Gemini with default thinking**	5-90 s per page	→ low thinking, 6 pages in parallel: ~5 s
× **Gemini cell by cell on any table**	over 10 min on one army list	→ at most 3 mismatches per table
× **Re-reading cells with qwen3.8 on the server**	33 s per army list, 0 corrections	→ moved to stage 2

Figure 12. What worked and what did not. All measurements come from real archive documents.

8. Limitations and open problems

- **No reference set yet.** I have not measured character-level accuracy. The figures in this report say how much the system doubts and how much it resolves, not how many errors remain unmarked. An error on which two readers agree, with no vision model contradicting it, goes through unmarked.
- **Classification is not validated.** On the 106 samples, the detected type matched the catalogue's in 33 cases. Many differences are debatable: the catalogue uses “other” for what the classifier calls an official gazette or an object photograph, and some files do not match their record. Even so, a systematic review is still missing. There are also real ambiguities between types, such as plate vs drawing or commission vs warrant.

- **Gemini's confidence is self-reported, not calibrated.** It is useful for ordering human review, but I have not checked that 0.9 means 90% correct.
- **Tables with “totals” that are not sums.** They are flagged for review rather than interpreted.
- **Human review capacity.** After a single night, the dictionary held about 3,000 pending entries. At the archive's scale it will need prioritising: by frequency, by confidence and by document type.
- **Dependencies.** If the second machine is off, there is no cross-review; if the Gemini API does not respond, there is no last resort. In both cases the document is still processed, and the result says what was missing.
- **Gemini cost.** It is logged per job (input and output tokens) but not yet measured at scale. A booklet page with one gap used about 2,400 input and 1,300 output tokens; the six manuscript pages, 7,638 and 21,078.

9. Next step: the 300,000 pages

This report is preliminary. Everything above was measured on about 4,000 pages; the archive has more than 300,000. At about 6.8 s per page, as in the production night, that would be about 570 hours of continuous processing — three to four weeks. Chaining the stages should reduce it; uncapped review and Gemini add to it. It is an order of magnitude that I will measure along the way. The plan is to start with a block of about 10,000 pages, representative of the archive's types, and use it to establish:

1. **A reference set** of about 200 pages transcribed or checked by hand, spread across types, to measure for the first time the real error rate, the rate of unmarked errors and the calibration of Gemini's confidence.
2. **Real speed and cost** per document type: pages per hour, time per stage and Gemini tokens per page.
3. **Human-review volume**: new dictionary entries per 1,000 pages, and how many the dictionary itself resolves as it grows.
4. **Stability**: failures and exceptions per 1,000 pages.
5. **The bottleneck**: if it is still table reading on the server, tackle it before running the rest.

With that data I will know whether the current setup is enough for the whole archive, or whether it needs more capacity, different thresholds or a change in which documents go straight to Gemini. The final version of this report will be written with those results, and the archive and catalogue will be published online as they are processed.